

Quote No OF13004048/4

Date 25.06.2025

SHORT QUOTATION / OFFER



FORM P 350

CUSTOMER INFORMATION

엠엔씨전자 비나

M&C Electronics Vina

강영규 Manager / Purchasing Team

Lot J1, Que Vo Industrial Zone(Expend area),

Phuong Lieu Commune, Que Va District

160000 Vietnam

Quote No OF13004048/4
Product FORM P 350

EXECUTIVE SUMMARY

GF Machining Solutions Co. Ltd
5F, Main building, K-tower,
88, Iljik-ro, Manan-gu, Anyang-si,
Gyeonggi-do, Seoul 13901

Prepared by :

Deion JH Hwang
GF Machining Solutions
Phone:
E-mail: jh.hwang@georgfischer.com



FORM P 350



Item	Description	Quantity
MACHINE		
995001175	FORM P 350	2 PC
VARIANTS		
EDMX.08	Console on rotative support (Attached to the machine frame)	2 PC
EDMX.13	Erowa ITS chuck	2 PC
DDX2.02	Standard C-Axis	2 PC
DDX2.01	Dielectric Agregat pump 60Hz (Countries using 60Hz)	2 PC
TOOL CHANGER		
D02.004	Automation kit for external robot (Erowa/3R)	2 PC
OPTIONS		
DDX2.32	LED light for workarea	2 PC
EDM.032	eConnectivity	2 PC
W28.021	SPD Surge Protection Device	2 PC
PACKING & DOCUMENTATION		
D22.002	Packing with wooden crate	2 PC

Quote No OF13004048/4
Product FORM P 350

Item	Description	Quantity
Contract type embedded with the machine		
CSP995011	Bronze - Recover fast	2 PC
Other service contracts		
200011529	My rConnect - Remote Access - Trial 1st year	2 PC
200011531	My rConnect - Live Data - Trial 1st year	2 PC
THIRD PARTY PRODUCTS		
Z100002	Air Dryer	2 PC
Z100002	Chiller	2 PC
Z100002	EDM Oil	4 PC
Z100002	Magnetic Table	2 PC
913746200	Sensor with ball Erowa Φ 2mm	1 PC
Z100002	Stabilizer	2 PC
SERVICES		
Z100003	Basic machine Installation	2 PC
Z100003	Operation Training	2 PC
Z100003	Transportation	2 PC

Quote No OF13004048/4
Product FORM P 350

CONFIGURATION TOTAL

ADDITIONAL EQUIPMENT AND SERVICES

Item	Description	Quantity
Z100002	Table reference ball 3mm	2 PC

OFFER PRICE

USD 310.000

Freight charges are quoted on the basis of current estimates. Additional charges may apply in case market prices change significantly until actual transport occurs.

Quotation valid to 23.07.2025

Quote No OF13004048/4

Product FORM P 350

TERMS AND CONDITIONS

Delivery

계약 후 3~4 개월 이내

본 계약에서 납기라 함은 계약 목적물을 "매수인"이 정하는 장소에 "매도인"이 인도하는 시점을 말한다. 단, 원자재 공급, 노사분규, 차량 운송 사고, 정부의 조치 등 "매도인"의 납품 지연 사유와 납품 장소의 완공 지연, 부대 시설 미비, 노사분규, 천재지변에 의한 "매수인"의 인수 지연 사유가 발생할 시 상호 협의하여 인도 시기를 연장할 수 있다.

"매수인"은 지연이 "매도인"의 잘못으로 인해 발생했으며 "매수인"이 그러한 지연으로 인해 손실을 입었다는 사실이 입증될 수 있는 한 지연된 배송에 대해 청산된 손해배상을 청구할 수 있다. 다만 고객에게 대체 자재를 공급할 수 있는 경우, 후자는 지연에 대한 손해배상을 청구할 수 없다. 배송 지연에 대한 손해배상은 매주 지연될 때마다 0.5%를 초과해서는 안 되며, 어떤 경우에도 지연된 공급물품 계약 가격의 5%를 초과할 수 없다. 지연의 첫 2주 동안은 어떠한 손해배상도 지급되지 않는다. 지연은 이 조항에 명시적으로 규정된 권리와 청구권 외에 고객에게 어떠한 권리와 청구권도 부여하지 않는다.

Prices

본선 인도 조건

계약물품의 인도방법은 "매수인"이 지정한 선박에 "매도인"이 약속한 화물을 적재, 본선상에서 화물의 인도를 마칠때까지로 한다.

Payment

계약금: 계약시 30%

중도금: 납품시 60%

잔금: FAC후 10%

"매수인"은 본 계약물의 인수 후 대금 지불 조건대로 대금 지불을 이행하여야 한다. 대금의 지불 수단이 어음인 경우, 그에 상응하는 담보를 "매도인"에게 제공한다. "매수인"의 대금 지불 및 수금에 대한 제반사항은 "매도인"이 수행하며, 은행자금 및 리스 자금으로 지불시에는 반드시 "매도인"에게 지불한다. "매수인"이 본 계약의 대금 지불, 일자 및 지불 조건을 이행하지 못하였을 경우, "매도인"은 "매수인"에게 연리 5%의 연체 이자를 청구할 수 있다.

무역 관세 정책의 불확실성과 관련된 GF Machine Solutions의 판매 조건 변경

"판매자는 정부의 조치(수입/수출 관련 비용, 관세, 기타 제세공과금의 변경을 포함하되 이에 한정되지 아니함)로 인해 견적상 상품 가격에 영향을 미치는 사정이 발생한 경우, 선적 이전에 해당 가격을 조정할 수 있는 권리를 보유한다."

Warranty

설치 후 12개월

본 계약물의 품질보증기간은 "매도인"이 계약 물품을 "매수인"에게 인도한 날로부터 12개월 동안이나 이 기간중 "매수인"이 과실이나 고의에 의한 하자 발생시 상호 협의하여 유상처리한다.

Quote No OF13004048/4

Product FORM P 350

Reservation of title

1항 : "매수인"이 본 계약물에 대하여 대금 지불 조건을 이행하지 않을 경우 "매도인"의 서면 통지로 계약은 해지되며 "매수인"은 즉시 계약 목적물을 "매도인"에게 반환한다.

2항 : "매수인"은 본 계약물의 대금 완제시까지 "매도인"의 승인 없이는 물품을 제3자의 채무 변제를 하기 위한 저당 설정 등 어떠한 형태로의 처분도 할 수 없다.

3항 : "매수인"이 주소를 이전하였거나 "매도인"의 채권 보전 등 계약 이행에 영향을 미치는 사유가 발생하거나 발생할 우려가 있는 경우 "매수인"은 이를 즉시 "매도인"에게 통보하여야 한다.

Limitation of Liability

계약 또는 그 위반과 관련하여 "매수인"의 클레임이 존재 하는 경우, 청구 총액은 각 구매 주문서에 따라 "매수인"이 지불한 가격으로 제한된다. 특히 손해배상, 가격 인하, 계약 해지 또는 철회에 대해 명시적으로 언급되지 않은 클레임은 제외된다. 어떤 경우에도 "매수인"은 공급품의 결함을 시정하는데 드는 비용에 대한 보상 외에 손해배상을 청구할 수 없다. 이는 특히 생산 손실, 사용 손실, 주문 손실, 리콜 비용, 이익 손실 및 기타 직간접적 또는 결과적 손해를 의미하지만 이에 국한되지 않는다. 이러한 책임 제한은 공급업체의 중대한 과실 또는 고의적 위법 행위로 인한 책임이나 공급업체의 행위 또는 부작위로 인한 사망 또는 신체 상해에 대해서는 적용되지 않는다.

Quote No OF13004048/4

Product FORM P 350

Other conditions of sales

1항 : "매수인"은 "매도인"이 직접 또는 "매도인"의 의뢰에 따라 행하는 신용조사에 성실히 응해야 한다.

2항 : 본 계약서에 유첨되는 서류의 기재사항은 본 계약서의 약정사항과 동일한 효력을 가지며 본 계약서와 중복되는 조항은 본 계약서의 조항에 우선한다.

3항 : 본 계약에 명시되지 않은 사항은 일반 상관례에 따른다. 본 계약에 관하여 소송의 필요가 생긴 경우 "매도인"과 "매수인"이 합의하여 관할 법원을 지정할 수 있다. "매수인"과 "매도인"은 본 계약의 당사자로서 신의를 성실하게 이행한다.

4항 : MAS 기계 활성화 시스템 공급자는 계약 합의된 구매 가격의 전액 지불을 고객의 가장 필수적 의무로 간주한다. 따라서 여기에 명시 합의된 지불 조건과 관련하여 전액 지불하지 않거나 그 일부를 지불하지 않는 것은 본 계약의 중대한 위반으로 간주된다. 고객은 장비에 기계 활성화 시스템이 장착되어 있음을 알고 동의한다. 이 시스템을 통해 공급자는 기계 기능의 전부 또는 일부를 일시적 또는 영구적으로 비활성화 할 수 있다. 고객이 지불 불이행 상태에 있는 경우, 계약서에 동의한 대로 시간이 지남에 따라 기계 기능의 전부 또는 일부가 비활성화되어 코드를 입력하지 않는 한 기계가 작동하지 않을 수 있다. 공급자는 고객의 금액 지불로 계약 위반이 해결될 때까지 기계 활성화를 일시 중지할 권리가 있다. 기계 기능의 전체 또는 부분 비활성화는 15 일 전에 기계 화면에 메시지가 표시 되어 코드를 입력하지 않으면 기계가 비활성화 된다는 것을 고객에게 알린다. 이 알림은 기계 기능의 전체 또는 일부가 비활성화되기 전에 적어도 두 번 표시되어 기계가 작동하지 않을 때까지 남은 일수를 고객에게 알린다. 고객이 공급자에게 서면으로 품질 불만을 전달한 경우, 공급자는 합의된 모든 사양을 충족할 때까지 기계가 전체 기능을 유지하도록 보장한다. 공급자는 고객 클레임에 대한 적절한 조사 후 그러한 클레임이 근거가 없다고 합리적으로 결론을 내릴 경우, 기계의 전체 기능 활성화는 연장되지 않으며, 대금 지불 기한은 이후 15 일 또는 최초 만기일 중 더 늦은 날짜로 적용된다. 고객이 관할 법원에 클레임을 제기하지 않는 한 적용된다. 공급자는 이러한 조건에 따라 기계가 자동으로 비활성화될 경우, 비용 손실 또는 손상에 대해 책임지지 않는다. 고객은 시스템을 제거, 비활성화, 수정 또는 변경할 권한이 없다. 이를 위반하면 기계 보증이 무효화된다.

러시아 또는 벨라루스로 재수출 금지 조약

비 EU 국가에 사업장이 등록된 구매자가 EU이사회 규정 제12조 833/2014호 또는 제8조 2024/1865호의 범위에 해당하는 경우 다음 사항이 적용됩니다:

- (1) 구매자는 본 계약과 관련하여 공급된 물품 중 EU이사회 규정 제 833/2014 제12g조 또는 2024/1865 제8g조의 범위에 해당하는 물품을 러시아 연방 또는 벨라루스에 직접 또는 간접적으로 판매, 수출 또는 재수출하거나 러시아 연방 또는 벨로루시에서 사용하기 위해 판매, 수출 또는 재수출 하는것은 불가합니다.
- (2) 구매자는 (1)항의 목적이 리셀러를 포함하여 상업 체인의 하위에 있는 제3자에 의해 좌절되지 않도록 최선의 노력을 기울여야 합니다.
- (3) 구매자는 리셀러를 포함하여 상업 체인의 하위에 있는 제3자가 (1)항의 목적을 좌절시킬 수 있는 행위를 감지하기 위한 적절한 모니터링 메커니즘을 설정하고 유지해야 합니다.
- (4) (1), (2), (3)항의 위반은 본 계약의 필수 요소에 대한 중대한 위반으로 간주되며, 판매자는 다음을 포함하되 이에 국한되지 않는 적절한 구제책을 모색할 권리가 있습니다:
 - (i) 본 계약의 해지.
 - (ii) 본 계약의 총액의 20% 또는 수출된 상품 가격중 높은 금액의 위약금.
- (5) 구매자는 (1), (2), (3)항을 적용하는 데 문제가 있는 경우, (1)항의 목적을 좌절시킬 수 있는 제3자의 관련 활동을 포

Quote No OF13004048/4

Product FORM P 350

함하여 즉시 판매자에게 알려야 합니다. 구매자는 판매자가 (1), (2) , (3)항에 따른 해당 의무 준수 정보를 요청하는 경우 2주 이내에 판매자에게 제공해야 합니다.

Quote No OF13004048/4
Product FORM P 350

MACHINE DETAILS



GF Machining Solutions is pleased to propose the AgieCharmilles FORM P 350 with PC numerical control, utilizing micro-processors that allow the automatic machining of openings, from roughing to finishing, by the automatic step-by-step execution of elementary machining operations, machining cycles, part programs, and any other necessary operations between programs, such as tool changing, filling up the work tank, etc.

- Space saving compact design
- High speed Z pulse 15 m/ min (49.2 ft/min) for no flushing applications with acceleration speed of 5 m/sec² (8.2 ft/sec²)
- Travers for X, Y, Z travel of 6 m/min (19.6 ft/min) for fast tool changing and measuring time
- 3 sided drop tank design for easy access
- 50 nm / 0.05 µm/ 0.000002" in resolution glass scales on all axes
- Collision protection on all axes
- AC FORM HMI - Windows PC based

Quote No OF13004048/4

Product FORM P 350

INCLUDED FEATURES

MACHINE

995001175 FORM P 350

Design

- Machine dimensions:
 - 1900 x 1690 x 2398 mm
 - 74.80 x 66.53 x 94.4 in
- Short, cast-iron C frame to ensure static and dynamic rigidity of the mechanical structure
- X, Y and Z axes are laid out above the main column
- This arrangement guarantees optimum accessibility and accuracy
- Machine equipped with a Z axe

Regulation

- Machinery directive 89/392/EEC
- EMC directive 89/336/EEC (EN 55011/Emission and EN 50082-2/Immunity)
- Low voltage directive 73/23/EEC

NB: For the machines sent outside Europe the regulation will not fully comply with the EMC norms but fully complies with safety regulation.

X, Y, Z axes

- Travel:
 - 350 x 250 x 300 mm
 - 13.78 x 9.84 x 11.81 in
- Maximum manual displacement speed (X, Y):
 - 6 m/min
 - 19.7 ft/min
- Maximum displacement speed (Z):
 - 15 m/min
 - 49.2 ft/min
- Maximum machining acceleration (X, Y)
 - 2.5 m/sec²
 - 8.2 ft/sec²
- Maximum machining acceleration (Z)
 - 5 m/sec²
 - 16.4 ft/sec²
- Axes equipped with precision linear rolling guide way increasing the movement accuracy and machining accuracy
- X, Y and Z equipped with loop servo control (Positioning control with glass scales and encoder) ensure pitch accuracy
- Linear displacement of all axes are inspected and software corrected using laser interferometer equipment according

Quote No OF13004048/4

Product FORM P 350

to the ISO 230-1 (geometric accuracy of machines) and ISO 230-2 (test code for machine tools - Determination of accuracy and repeatability).

- Automatic system provide optimum and permanent lubrication of the machine's vital components (PMC controlled)
- Spindle designed to receive a C axes, various chucks of EROWA, System 3R, HIRSCHMANN and MECATOOL

4 flushing systems, with pressure gauge, allowing the following possibility:

- 2 laterals flushing
- 1 workpiece injection flushing
- 1 electrode injection flushing
- 1 dielectric suction
- Continuous high-pressure injection, adjustable from 0 to 1.5 bars, lateral or through the workpiece
- Pulsed flushing injection synchronized with retraction of the electrode when pulsating

Machine table

- T slots for workpiece or fixture clamping
- The three sides drop tank allow easy access to the machining zone
- Dielectric level is programmable as well as the drop tank position
- Double check system stops the machine if the level of dielectric falls below the pre-set level or if there is an excessive rise in the temperature of the dielectric.

Front protection

Front protection can be slid up and down with the button located next to the remote control.

Dielectric unit

This includes:

- Filtration of dielectric
- Flushing
- Filling of dielectric
- Maintaining the dielectric level
- Cooling the dielectric level

4 integrated paper cartridge filter

Closed loop heat exchanger allows connection to separate cooling system to maintain dielectric temperature. Dielectric cooling circuit using heat exchangers built into the reservoir; automatic regulation of outside cooling fluid flow done by an accurate temperature sensor.

Flame Detection

The machine is equipped with a Flame Detection Unit (FDU) sourced by GFMS.

With sensors in the work tank, it allows the machine to suspend the erosion if a flame is detected.

(Compliance with ISO 28881:2022)

Electrical cabinet and power supply

The electrical cabinet incorporates:

- Pulse spark generator
- Numerical control windows

Quote No OF13004048/4

Product FORM P 350

- All the power of the machining center

Standard voltages available:

- 380 V
- 400 V

Voltage variation: +/- 10%

Micro breaks less than 4 ms (permitted)

Integrated battery back-up system guarantees memory retention for 100 hours without external power.

For other voltage supplies an autotransformer is necessary.

Remote control

2 types:

- With buttons (standard)
- With hand wheel (option)

Main functions are:

- Start and stop
- Axes controlled (movement, direction, and speed)

Touch screen with same function on console

Generator/Technology

- Intelligent Speed Power Generator (ISPG): 80 A (standard) and 140 A (option)
- Constant optimization of the orbiting movements for high efficiency planetary erosion with PLS (Predictive Learning System)
- Automatic optimization and maximum security of the EDM process with ACC/ACO
- Automatic selection and adaptation of the technology based on machining objectives and electrode data for a maximal removal rate and minimum electrode wear with TECFORM

SPAC

- Allows eliminating any short circuit by applying a specific discharge during a fraction of a second.
- Can be activated according to a configuration whatever the current value except during pulsation using capacitors.

Multi-task Numerical Control / Human Machine Interface (HMI)

- GFAC Homemade CNC unit is a high-performance multiprocessor Pentium with 2 Gigabytes of RAM which provides fast processing with a high level of reliability.
- AC FORM HMI offers clear choices for each application. The operator introduces—on a single screen—machining parameters such as surface state, depth, machining cycle, and type of application to extract the best generator performance.
- Allows multi-tasking allowing the operator preparing the next job to regain control of the machining by changing a page with a simple pressure on the screen icon or menu.
- Programs and data files are stored into a 60 Gigabytes hard drive
- The control is equipped in standard with
 - Ethernet Board
 - Port RJ 45

Quote No OF13004048/4

Product FORM P 350

- 2 USB ports
- Communication protocols available are:
 - TCP/IP
 - IPX
 - NetBEUI

All the following steps are available during machining:

- **Job**: create new job or re-using an existing one or use job model
- **Tools**: describe the number of electrode, the under-sizes, and the offset
- **Parts**: information about the workpiece such as, quantity, references, workpiece high
- **Cavities**: information about the location of a single or multi-cavity (definition can be done by using pre-defined patterns)
- **EDM**: All information regarding the machining is summarized on a single screen to automatically generate the machining strategy
- **Sequence**: define in an easy way the complete machining process, which electrode will be used and for which cavities. Also, the graphic simulation is available to check a job before executing the program
- **ISO**: Based on all the above information, the CNC will generate the complete ISO program including generator setting, tool change, positioning of the cavity...

Large choice of additional functionalities

Measuring cycles for tooling:

- **REFERENCE BALL**: This cycle is used to define one reference point in the space that will be used to measure the tools offsets
- **TOOL ALIGNEMENT 1**: This cycle gives the possibility to find out the angle of the electrode by measuring on one surface. This offset will be taken into account on each motion of the C axis
- **TOOL ALIGNEMENT 2**: This cycle gives the possibility to find out the angle of the electrode by measuring on Z and one surface. This offset will be taken into account on each motion of the Z and C axis
- **TOOL OFFSET 1**: This cycle is used to define the tool offset in X, Y, Z. All these offsets will be automatically loaded into the offset table
- **TOOL OFFSET 2**: Same as above cycle, except that this cycle is used for dissymmetric electrode that needs more travel on one axis as on the other one

Measuring cycles for the work piece:

- **EDGE**: Measuring one edge of the work piece in Z, X or Y-axes or 3D measuring vectors
- **CENTERING**: Used to define the centre of a hole or a cavity on 1 or 2 axes (X, Y)
- **EXTERNAL CENT 1**: This cycle is used to find the centre of a work piece by measuring on the external surface
- **EXTERNAL CENT 2**: Same as the above cycle, except that in this case, a cavity is already located on the surface and need to offset the first probe (i.e. X, Y or Z)
- **CORNER**: Used to define the corner of a work piece
- **PART ALIGNEMENT 1**: This cycle gives the possibility to find out the angle of work piece by measuring on one surface. This value will be taken into account for all motions of the axes X, Y, Z and C
- **PART ALIGNEMENT 2**: This cycle gives the possibility to find out the angle of work piece by measuring on Z and one surface. This value will be taken into account for all motions of the axes X, Y, Z and C. 2
- **HOLES ALIGNEMENT**: This measuring cycle offers the possibility to define the angle of the work piece by measuring inside two holes

Quote No OF13004048/4

Product FORM P 350

- 3 POINTS INT. CENT: Measuring the internal centre of a hole by probing on 3 points
- 3 POINTS EXT. CENT: Measuring the external centre of a work piece by probing on 3 points

Machining cycles:

All these cycles have a specific ISO code which is completely transparent for the operator during the programming.

- DOWN: Plunge cut machining on one, two or three axes simultaneously
- ORB: This cycle combines the plunge and the orbital translation on a 45° cone
- EXPAN: This cycle plunge and a plane orbital translation
- ISOGAP: Equidistant gap machining along X, Y, Z
- CYLINDER: Combine plane orbital translation and plunge
- CONE: Conical machining opening upwards or downwards under an angle between 0-90° and around X, Y or Z-axis
- DIAG: This cycle enables the machining in the diagonal of a cavity. The operator can choose the number of diagonal as well as the angle of the translation
- HELIC: Helical machining with C-axis (option)
- TRAJECTORY: This cycle allows to define a trajectory of a profile by defining each segment (X, Y, Z, C) or (X, Y, Z, contouring mode)

N.B.: All the above machining cycles (except Helic) are index able in any direction in the space (3D)

Execution

This module is dedicated for operation's linked to the sparking of cavity:

- CONFIGURATION: Operator can configure the way to execute a program and make a simulation.
- GAP: Where all the sparking parameters are located (M, C, V, P, A, VPULS, T, B, R, U, SV). The operator has the possibility to modify these parameters of the current program. Also information of the sparking conditions is displayed as well as a graph of the removal rate.
- FOLLOW-UP: Overview of the ISO program in progress as well as generator settings. Visual graphic during machining.
- SEQUENCES FOLLOW: Displays the machining sequence (actual + coming sequences)
- HISTORY: Each generator setting executed will be stored in table with some indications about execution time, sequence references, machining cycle, machining time, etc.
- JOB LIST: Organize Job sequence according to manufacturing priorities
- PART EXPRESS: Allow interrupting the operation in progress in order to insert a more urgent Job.

Execution - Additional functionalities

- Dielectric management: Operator can set manually the flushing during machining
- Manual movement (X, Y, Z and C axes): move manually axes individually or simultaneously.
- Other: spindle on/off, retract the electrode from the cavity or contour and electrical touch on/off.

NB: Jobs described or executed are listed on a file. The operator can visualize the description of the file, modify or execute a Job. Operator can also use a Job as model and the explorer gives you the opportunity to copy, delete and rename the file.

Connectivity:

MT Connect is set by default. This option enables machine communication and allows publication of internal machine

Quote No OF13004048/4

Product FORM P 350

information and status.

VARIANTS

EDMX.08 Console on rotative support (Attached to the machine frame)

Console on rotative support (Attached to the machine frame)

EDMX.13 Erowa ITS chuck

- Mounted ex work
- ER-010591 1 pcs Uniholder
- 1 pcs Chucking spigot
- 1 set of 5 pcs insert plates
- Tolerances within 0.002mm
- Air blast cleaning of the Z - reference
- Flushing connection

DDX2.02 Standard C-Axis

This C-axis (without chuck) is incorporated into the Z-axis working head and is compatible with EROWA, System 3R, MECATool and HIRSCHMANN tooling systems. The tooling system must be specified with the order.

It has three main functions:

- Rotating an electrode, with a permanent speed, selected from 1 - 100 rpm.
- Indexing the electrode to a certain angle programmed in the NC.
- Controlled machining axis, usually operating together with one or more linear axis (simultaneous control), e.g. for machining a spindle shape (machining mode HELIC).
- Provision, for central dielectric flushing.
- Electrode clamping is made by pneumatic and can be activated manually or automatically.

Technical specifications:

- Maximum electrode weight with C-axis: up to 50 kg (depends on the spindle chuck)

For both static and dynamic use:

- Maximum electrodes inertia:
 - 1500 kg x cm²
- Measuring resolution:
 - 0.1 mdeg
- Rotation speed range:
 - 0.2 to 100 rpm
- Maximum torque:
 - 8 Nm
- Indexing resolution:
 - 0.001°
- Maximum machining current (rotating and static):
 - 140 A

Quote No OF13004048/4

Product FORM P 350

DDX2.01 Dielectric Agregat pump 60Hz (Countries using 60Hz)

Dielectric Aggregate pump 60Hz (Countries using 60Hz)

TOOL CHANGER

D02.004 Automation kit for external robot (Erowa/3R)

Each machine which will be equipped with an external robot (except RTC) needs to have this Set for Robot. This Set includes the cables, connectors, and software for the control of the RS 232 link with the robot.

OPTIONS

DDX2.32 LED light for workarea

EDM.032 eConnectivity

This powerful function is the answer to any demand of connectivity of the machine to a remote software. It allows the machine connected to the local network of a workshop to:

- Be controlled by a remote software.
- LOAD a job, START SUSPEND ABORT a job execution, send a JOB_EXECUTION_REPORT (Transfer the machining report at the end of the job execution), REFRESH tools or parts offsets and magazine position (that allows starting an execution without all tools + parts ready).
- Send information on its status to a supervisor.
- Machine identification, run, suspended state, current job in execution, current alarm, current part, current Wire
- Send information on all events happening on the machine.
- Job execution start-end, new tool load, new part load, tool to be loaded, part to be loaded, machine power failed, emergency Stop ON, OFF
- Send information on consumable status, and machining efficiency.

These data are used to track the machining process, performances, and running time.

This function supports the MT Connect protocol.

W28.021 SPD Surge Protection Device

To reinforce the protection against the damaging effects of transients caused by lightning, utility switching, internal load switching and more. In addition to the mains filter with integrated protection introduced as standard, an optional group containing a Type 2 Surge Protection Device (SPD) and related wiring.

PACKING & DOCUMENTATION

D22.002 Packing with wooden crate

The machine is fixed on a wooden transport. The material of four lateral plates and the roof is OSB (Oriented Strand Board).

Contract type embedded with the machine

CSP995011 Bronze - Recover fast

Bronze is the way to get faster assistance through My rConnect.

You are guaranteed 1 hour callback on support requests, 24/5 access to expert technicians, and fast track service intervention scheduling to ensure quick diagnostics, uninterrupted production and efficient troubleshooting.

Quote No OF13004048/4

Product FORM P 350

BRONZE

CSP000024 Yearly base contract

CSP000008 Fast track for support

Fast response time through My rConnect for digital service requests within 60 minutes.

Fast service engineer dispatching for onsite interventions.

CSP000012 24/5 expert support through My rConnect

Remote support through My rConnect can be reached 24 hours a day, 5 days a week (Monday to Friday)

Other service contracts

200011529 My rConnect - Remote Access - Trial 1st year

Remotely access to your machine anytime, anywhere, to monitor its status and optimize your workflow via the My rConnect web platform. Additionally, you can authorize GF Machining Solutions technical support to troubleshoot your machine remotely. This secure service is available through an annual subscription.

200011531 My rConnect - Live Data - Trial 1st year

Stay informed about your machine status anytime, anywhere. Monitor machine status and operation progress, receive notifications for status changes, messages, or alarms, and react quickly without needing to visit your plant. Combined with Remote Access, you can also view your machine's user interface remotely. This feature is available through an annual subscription.

THIRD PARTY PRODUCTS

Z100002 Air Dryer

Z100002 Chiller

Z100002 EDM Oil

Z100002 Magnetic Table

913746200 Sensor with ball Erowa Ø2mm

Z100002 Stabilizer

SERVICES

Z100003 Basic machine Installation

Z100003 Operation Training

Z100003 Transportation

Quote No OF13004048/4

Product FORM P 350

ADDITIONAL EQUIPMENT AND SERVICES

Z100002 Table reference ball 3mm

Quote No OF13004048/4

Product FORM P 350

SPECIFICATIONS

Machine		
Architecture	C-frame/Fixed table/Drop tank	
Dimensions (width x depth x height)	1900 x 1690 x 2398 / 74.80 x 66.53 x 94.4	mm / in
Total weight (without dielectric)	2800 / 6173	kg / lbs
Floor space (width x depth)	1900 x 1690 / 74.80 x 66.53	mm / in
Complies with "Machines, Safety and Health" directive	89/392/CEE	
Complies with "Electromagnetic, Compatibility" directive	89/336/CEE	
X, Y, Z axes		
X, Y, Z travel (width x depth x height)	350 x 250 x 300 / 13.78 x 9.84 x 11.81	mm / in
X, Y, axes speed	6 / 19.7	m/min / ft/min
Z axis speed	15 / 49.2	m/min / ft/min
Positioning resolution X, Y, Z	0.05 / 0.000002	µm / in
Work area		
Worktank size (width x depth x height)	800 x 520 x 350 / 31.49 x 20.47 x 13.78	mm / in
Worktable size (width x depth)	500 x 400 / 19.68 x 15.75	mm / in
Distance floor to clamping level	1000 / 39.37	mm / in
Min./Max. distance between table and chuck (With C-Axis Eco + System 3R Macro)	150/450 / 5.91/17.72	mm / in
Workpiece and electrode		
Max. electrode weight	50 / 110.23	kg / lbs
Max. workpiece weight	500 / 1102.31	kg / lbs
Max. workpiece dimensions (width x depth x height)	790 x 530 x 275 / 31.10 x 20.87 x 10.83	mm / in
Bath level (programmable)	100 - 305 / 3.94 - 12.01	mm / in
Dielectric unit		
Capacity	410 / 108.65	l / gal
Number of filter elements and type	4 Paper filter	
Generator		
Generator type	Intelligent Speed Power Generator (ISPG)	
Max. machining current (option)	80 / 140	A
Best surface finish	0.08	µm Ra

Quote No OF13004048/4

Product FORM P 350

Electrical supply standard		
Standard voltage	3 x 380V/400V $\pm$ 10%. 50/60Hz (50Hz standard)	
Cooling		
Heat exchanger dielectricum/water for the dielectricum	Integrated	
Control Unit		
Operating system	Windows	
Data input	15" LCD screen, mouse or touch screen, keyboard and remote control	
User interface	AC FORM HMI	
Expert systems	TECFORM	
Console support	Movable on cabinet or on foot	
Modules		
Z axis (15 m/min)	Standard	
Linear tool changer (option)	4 (Std. tooling), 5 (Combi tooling)	
Rotary tool changer (option)	16-80 pos. (Std. tooling), 32-160 pos. (Combi tooling)	
Flushing Injections	2 Laterals, 1 through the piece, 1 through the electrode, 1 suction	
Heat exchanger air/water for the cabinet	Option	
Multicavity flushing 6 outputs	Option	
3D probe measuring system for Erowa or System 3R	Option	
iQ graphite and copper module for reduction of electrodes' wear	Standard	
Standard C-Axis (option)		
Max. electrode weight on automatic chuck	25 / 55.12	kg / lbs
Rotation speed	0-100	rpm
Max. inertia	2000 / 683	kgcm ² / lbsin ²
Accura C-Axis (option)		
Max. electrode weight on automatic chuck	25 / 55.12	kg / lbs
Rotation speed	0-100	rpm
Max. inertia	5000 / 1700	kgcm ² / lbsin ²

My rConnect

Digital services for all your GF machines, anytime, anywhere

My rConnect is your gateway to advanced digital assistance. This innovative web-based platform provides seamless access to a wide range of digital services to boost machine uptime and productivity. Enjoy simple navigation and direct access to GF support, for all connected and unconnected machines, regardless of their age.

Comprehensive and free-to-use asset management tool for all GF machines

 My rConnect



Organize your workshop and facilities, and manage user access



Use chat, video conferencing, whiteboard, and file transfer for rapid interaction and problem-solving



Manage and access the complete history of service cases for improved decision-making



Easily access machine user manuals in one place, and upload additional documentation

My rConnect

Connect your machine for greater responsiveness and operational efficiency

Get first-class support and real-time insight by subscribing to Remote Access and Live Data service offerings, available on-demand for eligible machines.

Do you want to increase your equipment uptime?



Remote Access

- Connect and share files to/from your machine from anywhere to solve issues efficiently
- Get expert troubleshooting from GF with remote access to your machine's HMI

Do you know if your machine is operating now?



Live Data

- Increase your responsiveness to unexpected events with real-time status notifications and instant alerts.
- Track machine connection and job status anywhere, anytime

Exclusive support when you need it most

Combining Remote Access and Live Data with Success Packs enhances responsiveness and optimizes uptime.

For more information, contact your local Sales Representative.

Bronze

Silver

Gold